

TetriC

A C-based console Tetris game with
network multiplayer using socket programming

STUDENT

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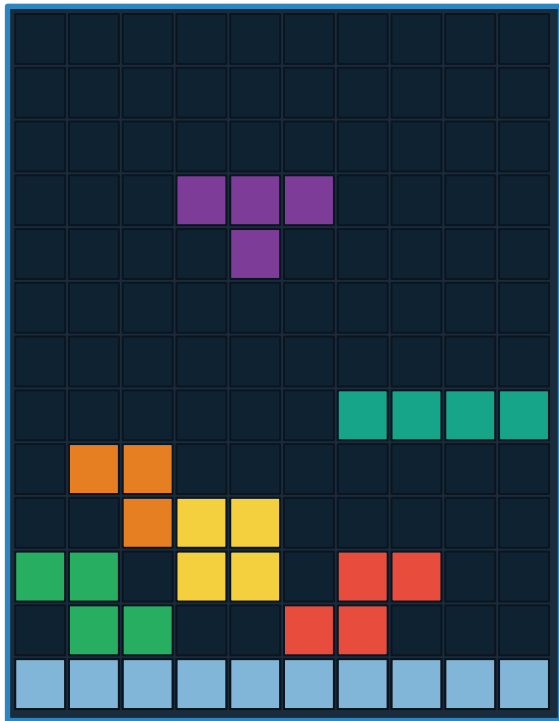
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What is TetriC?

- Inspired by the classic Tetris game
- Built entirely in C as a console application for Windows
- Runs in the terminal — no graphics library needed
- Supports both Local play and Network (online) play
- Developed as part of the Software Project Lab course

TETRIC



Console-based · Network-ready · Written in C

How the Game is Played

Goal

Rotate and place falling tetrominoes to fill complete horizontal lines and clear them from the board.

Scoring & Levels

Each cleared line earns points. Clearing multiple lines at once scores more. Speed increases as the level rises.

Losing Condition

If pieces stack up and reach the top of the board, the game ends. Last player standing wins in multiplayer.

Controls

← →

Move piece left / right

↑

Rotate piece

↓

Soft drop

Space

Hard drop

P

Pause game

Q

Quit game

Game Modes & Features



LOCAL MODE

- Single Player — play solo with scoring
- Two Player — side-by-side on the same keyboard
- Difficulty levels (Easy / Medium / Hard)
- Next-piece preview
- Per-player high score tracking

VS



NETWORK MODE

- Single Player over network
- Two Player over network
- Multiplayer — many players from different devices
- Server-client socket architecture
- Play over LAN or localhost

Common Features

Difficulty levels

Next-piece preview

Pause / Quit

Scoring system

Speed progression

High scores

Socket Programming & Multiplayer

- The game uses TCP sockets for real-time communication between devices
- Any number of clients can connect to a central server
- Server assigns a unique Player ID to each connected client
- Each client runs their own game loop and sends state to the server
- Server relays game state to all other clients and announces the winner
- Match ends when only one player remains active

Tested on localhost (127.0.0.1) and across LAN



Technical Structure / System Architecture

TetriC.c

Main Loop & Menu

Entry point. Handles main menu, mode selection, and game initialization.

Game.c

Config & Winner Logic

Manages difficulty settings, player configuration, and winner detection.

Board.c

Piece Movement & Line Clear

Core gameplay: tetromino placement, rotation, collision detection, line clearing.

UI.c

Console Rendering

Draws the game board, score display, and next-piece preview to the terminal.

Network.c

Socket Communication

Handles all TCP socket logic: server loop, client connection, state sync.

Globals.c

Shared State

Holds global variables and structures shared across all modules.

Complexity & Challenges Faced

⚡ Challenge

Fixed 2-player → Scalable multiplayer

Synchronizing game state over sockets

Echoed state packets causing flicker

Handling disconnections mid-game

✓ Solution

Refactored the player model to use an array of player structs with dynamic state, removing hardcoded player-count assumptions.

Designed a structured packet format. Each client sends its board state; server relays to all others every tick.

Added a sender ID in each packet so clients discard their own echoed packets and re-render only on remote updates.

Server marks disconnected players as inactive; remaining players continue and the last active player wins.

Testing & Validation

GAMEPLAY

- ✓ Single-player game loop runs correctly, pieces fall and lines clear
- ✓ Scoring increments correctly; speed increases with level
- ✓ Pause and quit keys work in both local and network modes

MULTIPLAYER

- ✓ Server starts, listens for clients, and assigns Player IDs
- ✓ Multiple clients (3+) connect via localhost (127.0.0.1) simultaneously
- ✓ Game state is correctly relayed; all clients see the same boards
- ✓ Disconnected players are removed; match continues for others
- ✓ Winner announcement is broadcast to all remaining clients

Testing environment: Multiple PowerShell terminals on Windows, using
localhost (127.0.0.1)

Learning Outcomes & Project Value



C Programming

Pointers, structs, memory management, and modular code design



Game Loop Design

Tick-based loop, state updates, and real-time input handling



Console UI

Terminal rendering with cursor control and flicker-free drawing



Socket Programming

TCP sockets, server-client architecture, send/recv loops



Data Structures

Arrays, 2D boards, structs for players, pieces, and game state



Debugging Skills

Multi-terminal testing, packet inspection, and edge-case fixing



System Design

Separating concerns: logic, UI, network, config in distinct modules



Distributed Thinking

Designing a game that stays consistent across multiple clients

Conclusion

- TetriC is a fully functional, console-based Tetris game written in C
- It goes beyond a basic single-player game by implementing real network multiplayer using TCP socket programming
- Players across different devices can connect to a central server and compete in real time
- The project demonstrates strong C programming skills alongside key concepts in game design, system architecture, and networked applications
- All features — scoring, difficulty, high scores, pause, and winner detection — are fully implemented and tested

Thank You!



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<https://github.com/Shourov735/TetriC>